higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

LOGIC SYSTEMS N6

30 JULY 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 True – It uses comparator op-amps and output is instantaneous.

1.2 True – Bit density has increased over the last twenty years. This has made it difficult to be exact about the correctness of different classifications. With millions of gates on one chip Intel Pentium microprocessors fit into either of the two following classifications:

VLSI: Very-large scale integration
LSI: Large-scale integration

1.3 True – A modem (acronym for modulator/demodulator) interfaces a computer with a telephone or wireless system to use e-mail and connect to the internet.

1.4 True – A USART (universal synchronous/asynchronous receiver/transmitter) forms part of the modem card. It converts serial data (originally from the telephone line) to the parallel format used by the computer. It also converts parallel to serial or operates in serial mode.

1.5 True – The operational amplifier is configured as a summer.

1.6 True – The purpose is disconnection (isolate) from the bus.

1.7 True – A port is a common interface.

1.8 False – There is no significant difference in stack use.

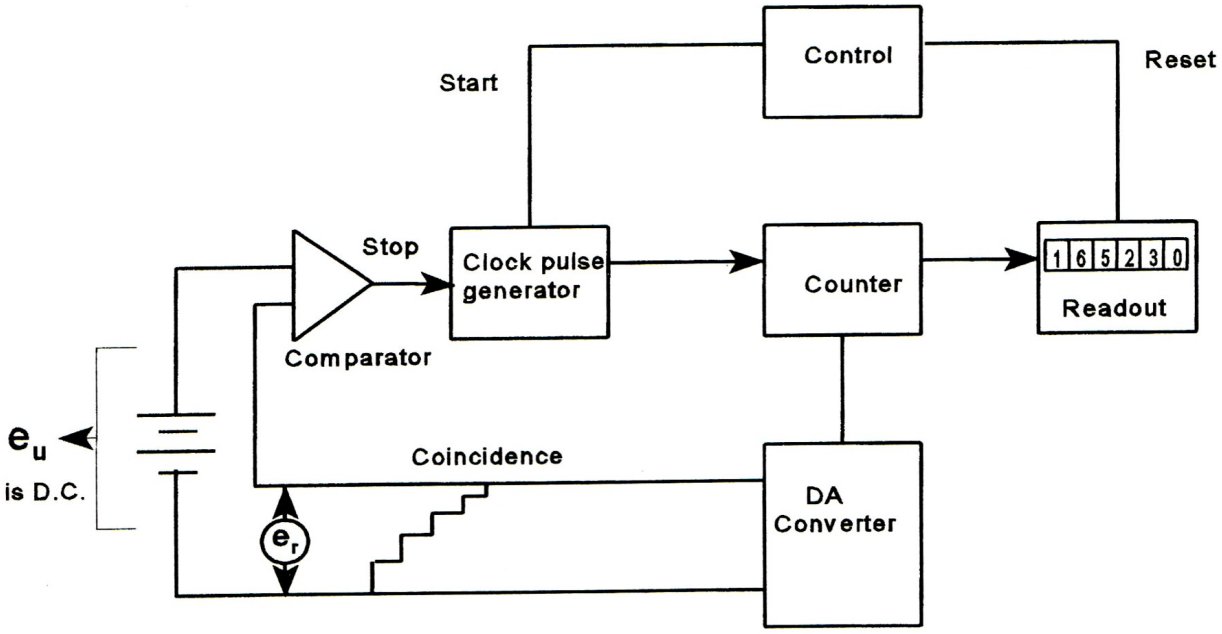
1.9 True – The ASCII code became a worldwide standard in the 1960s because of IBM's dominance of the computer market.

1.10 False – A keyboard is not an interface; it is an input device.

(10 × 2) **[20]**

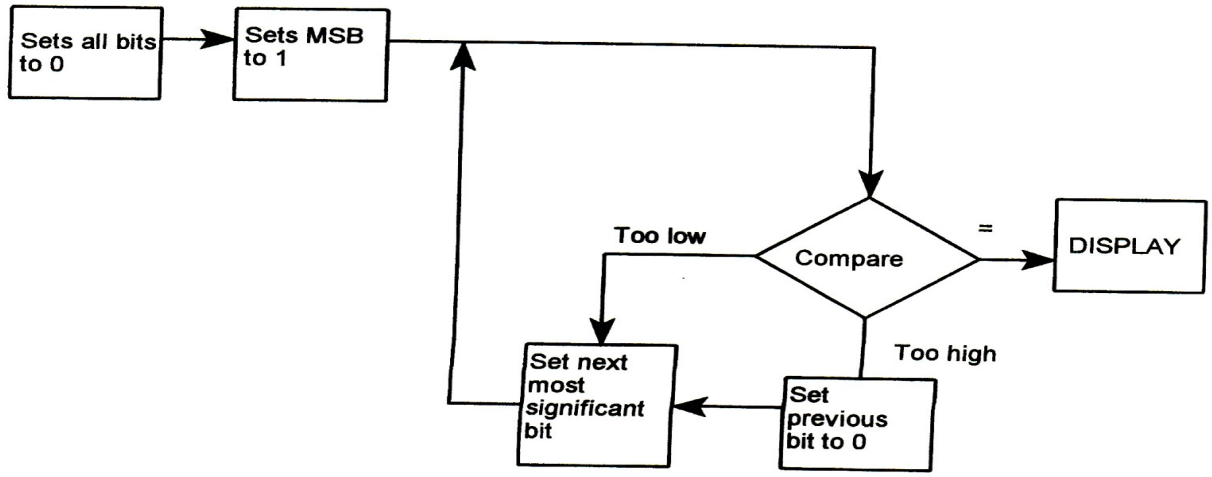
QUESTION 2

2.1



(10)

2.2



(9)

2.3 Assume a six-bit converter:

Clear register	000000✓	Binary input equivalent = 001000✓
Set MSB	100000	Too high – clear bit and set next one✓
Reset and set next MSB	010000	Too high – clear bit and set next one✓
Reset and set next MSB	001000	Equality✓
Equality to input	001000	Equal to input so stop and display✓

(6)
[25]

QUESTION 3

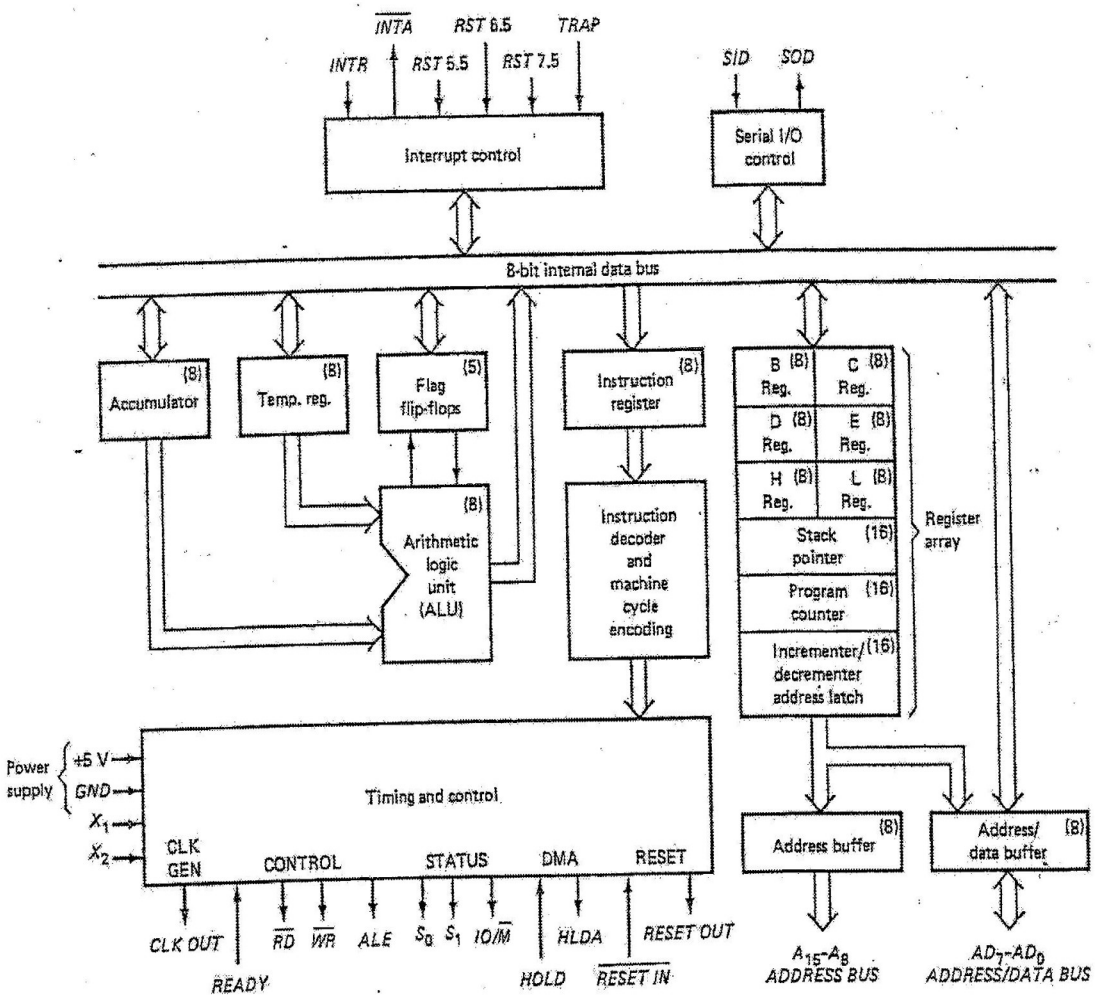
3.1 **Von Neumann architecture:**

- It has only one memory, namely RAM which is accessed twice during each machine cycle.
- During the fetch routine the ram is accessed to fetch the instructions routine into the instructions register.
- During the execute routine the RAM is accessed to get the operand out of the operand address.

Harvard architecture:

- It has two separate memories, namely RAM and ROM.
- One memory is usually EPROM or EEPROM which stores the program instructions permanently.
- The RAM stores only the operands.
- The fetch and execute routines will access two separate memories. (10)

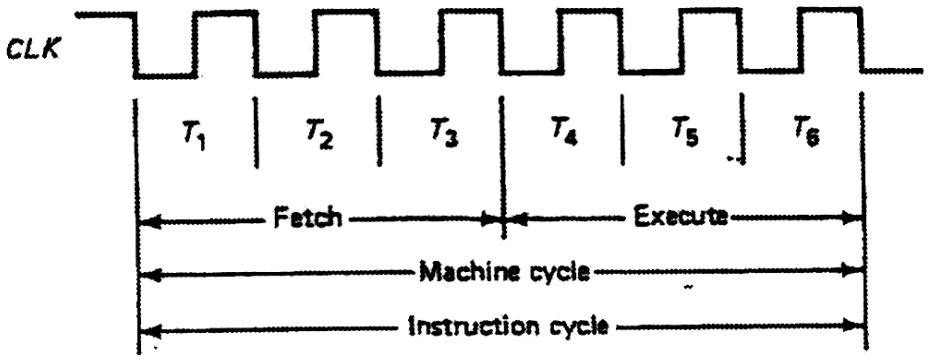
3.2



(10)
[20]

QUESTION 4

4.1 The cycle (microinstruction cycle) in the control unit that carries out a program instruction (like load)



(a) (6)

4.2 4.2.1 Temporary data stored during subroutines and interrupts (1)

4.2.2 Subroutines and interrupts (1)

4.2.3
- The stack stores active register data and the contents of the program counter (PC).
- On the return instruction this data is returned to the original registers so that the main program can be resumed. (2)

4.3 **PROGRAM**

$X = U^2 - t$
x = u.u – t

INSTRUCTION LOCATION	INSTRUCTION WORD		DESCRIPTION
	OP-CODE	ADDRESS	
BCCA	LOAD	F1B4	Load u✓
BCCB	MULT	F1B4	Multiply by u✓
BCCC	BRAN	BCD1✓	Branch if Acc. < 0✓
BCCD	SUBT	F1B3	Subtract t✓
BCCE	BRAN	BCD1✓	Branch if Acc. < O✓
BCCF	STOR	F1B7	Store in x✓
BCD0	OUTP		Output✓
BCD1	HALT		Halt✓

NOTE: Other programming sequences are also accepted. Either u or t can be loaded first. (10)
[20]

QUESTION 5

5.1 The gain of a noninverting operational amplifier must always be greater than one. In this circuit the gain is less than one and therefore the operational amplifier is faulty and must be replaced.

INPUT ✓	×	GAIN✓	=	OUTPUT
12 volts	×	0,5 ✓	=	6 volts

Gain = 0,5: which is less than one. So: Change the op amp. (6)

5.2

ADVANTAGES	DISADVANTAGES
• Only two resistor values needed✓✓ • Range of the converter is easily extended✓ • Nonstandard resistors can be used✓✓	• Cannot convert BCD✓

(6)

5.3 To improve the resolution of a D/A converter the number of bits in the register must be increased. (3)

[15]

TOTAL: 100

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